drift speed = ... m s^{-1} [2]

5 **(a)** A particle has mass m, charge $+q$ and speed v.

State the magnitude and direction of the force, if any, on the particle when the particle is travelling along the direction of

(i) a uniform gravitational field of field strength g,

.....

..... [2]

(ii) a uniform magnetic field of flux density B.

.....

..... [1]

(b) Two charged horizontal metal plates, situated in a vacuum, produce a uniform electric field of field strength E between the plates. The field strength outside the region between the plates is zero.

The particle in **(a)** enters the region of the electric field at right-angles to the direction of the field, as illustrated in Fig. 5.1.

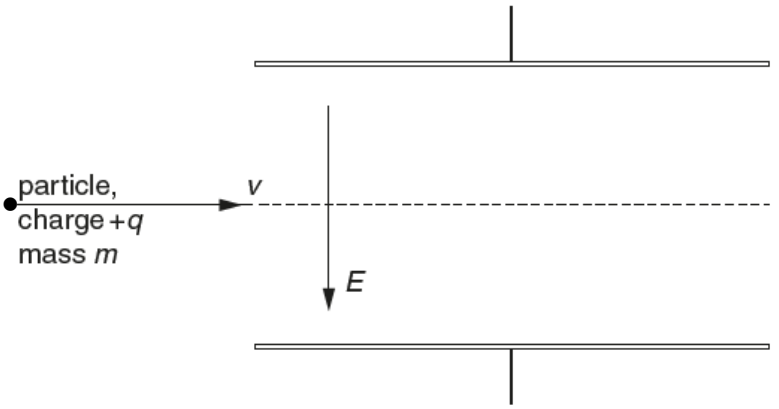

Fig. 5.1

A uniform magnetic field is applied in the same region as the electric field so that the particle passes undeviated through the region between the plates.

(i) State the direction of the magnetic field.

.....

..... [1]

(ii) Derive, with clear explanations, an expression for the speed v in terms of the magnitudes of the electric field strength E and the magnetic flux density B.

[2]

(c) The same particle in (a) now enters a non-uniform magnetic field B at an angle θ with the horizontal as shown in Fig. 5.2.

By considering the components of the velocity parallel to and at right angles to the magnetic field, explain the subsequent path of the charged particle in the field.

Draw a sketch to illustrate the path in Fig. 5.2.

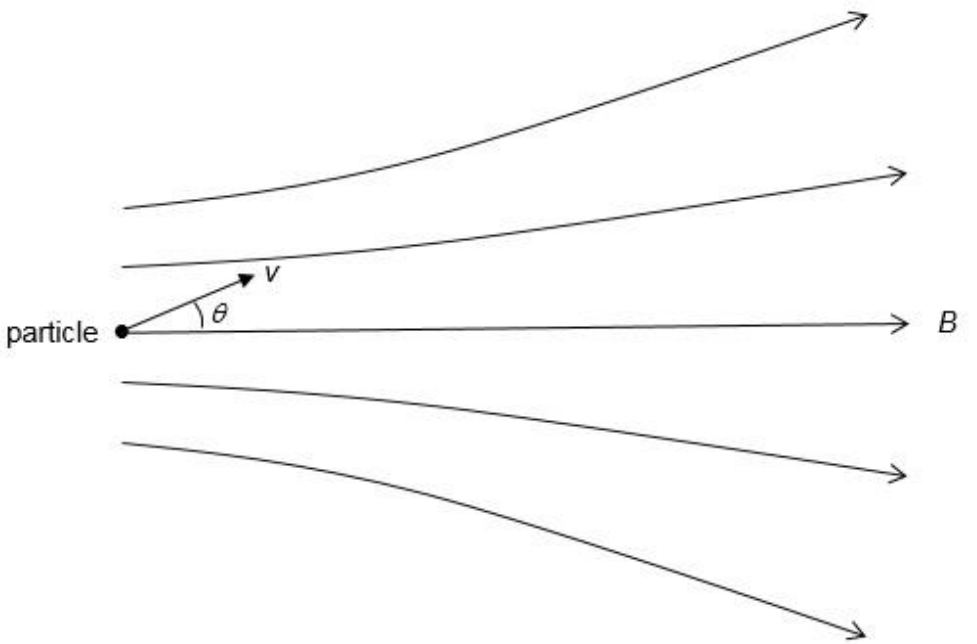

Fig. 5.2

.....

.....

.....

..... [4]